

MITHUUSAN KIRUPANANTHAN

Software Engineer | Full-Stack, Backend, and AI-Enabled Applications

647-568-7527 | mithuusank@gmail.com | LinkedIn | GitHub | Portfolio

Education

Toronto Metropolitan University

Sep 2021 – Apr 2025

Bachelor of Science (Honours), Computer Science

Toronto, ON

- **Coursework:** Algorithms, Data Structures, Databases, Operating Systems, Networks, Software Engineering

Work Experience

SGMC Canada

Jan 2026 – Present

Software Developer (Contract)

Toronto, ON

- Owning development of Connect2Talents' initial web application using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, building reusable page layouts across **10+ screens** and implementing sign-up/login, onboarding, protected dashboard, and document review workflows.
- Developing **Express.js** and **PostgreSQL** backend services with authentication APIs, protected routes, **role-based access**, and **AWS S3** file-upload workflows to support secure onboarding and document handling.

Muia Consulting

Sep 2025 – Dec 2025

Software Engineer Intern

Toronto, ON

- Built a **FastAPI** backend using **Google Gemini 2.0 Flash** and **Google Document AI** to convert unstructured tax and project documents into structured **SR&ED** claim data.
- Designed a resilient document-processing pipeline with OCR extraction, **Gemini** fallback, and normalized **JSON/CSV** outputs, reducing SR&ED workflow time by **50%** by eliminating manual document parsing and data entry.
- Developed backend services in **Python** to process timesheets, invoices, materials, and equipment records, calculate **ITC** eligibility and credit amounts, and enforce validation checks for accurate financial outputs.
- Implemented **Pydantic**-validated APIs and **Google Drive OAuth 2.0** integration for reliable file ingestion.

Stephen Lewis Foundation

Oct 2024 – Dec 2025

Program Officer (Data & Analytics)

Toronto, ON

- Delivered a **Power BI dashboard** and automated data intake workflow that cut manual reporting time by **50%**, giving leadership real-time access to consolidated metrics across **5 teams** in a **100+ employee** organization.
- Built and maintained **SQL**-based data pipelines with validation and transformation processes, replacing a days-long reporting process with instant access to decision-ready insights.

Equitable Bank

May 2022 – Dec 2022

Cloud & DevOps Engineer Intern

Toronto, ON

- Designed and implemented **CI/CD pipelines** to automate build, test, and deployment workflows, supporting **25+ production deployments** for the digital banking card project and reducing manual release effort.
- Built an **API dependency map** for the digital banking card project to help the team identify downstream service impacts faster and reduce downtime risk during weekly and monthly releases.

Relevant Projects

NovaPrep | React, Express.js, AWS Nova, AWS Bedrock, AWS Amplify

- Built and deployed **NovaPrep**, a live AI interview platform, using a React/Vite frontend and Express.js backend to support conversational mock interviews for 50+ concurrent users.
- Integrated **Amazon Nova Lite** and **Nova Sonic** through Amazon Bedrock for adaptive question generation, voice-based interviewer interactions, speech transcription, and structured answer evaluation.
- Implemented resume-aware personalization and rubric-based feedback pipelines, then deployed the application on **AWS Amplify** and **AWS App Runner** with automated CI/CD.

ForgeFit | React Native, TypeScript, Supabase, GPT-4o, USDA FoodData Central

- Built a cross-platform fitness application using **React Native** and **TypeScript**, delivering workout tracking, health management, and food logging features across **iOS** and **Android**.
- Implemented AI-powered meal analysis using **GPT-4o** and the **USDA FoodData Central** dataset to provide personalized nutrition insights with approximately **95% macro accuracy**.
- Integrated **Supabase** for authentication, session management, and cloud data storage, enabling secure user access and reliable persistence of workout and nutrition data.

Skills

Languages: Java, Python, TypeScript, SQL

Frameworks: React.js, Next.js, React Native, Tailwind CSS, Vite, FastAPI, Express.js

Tools/Cloud: Git, Docker, Jenkins, Linux, Power BI, PostgreSQL, Supabase, AWS (S3, App Runner)

AI/ML: OpenAI API, Google Document AI, Google Gemini, AWS Bedrock